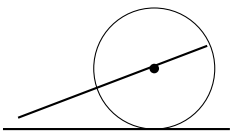
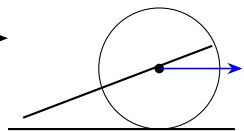
碰前定轴

碰后解锁

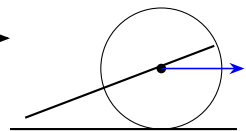
首次纯滚动



碰撞



滑动



$\omega_f > 0, \quad v_O = 0$

$v = v_0, \quad \omega_w = 0$

$v = v_1 = r\omega_w$

$u = v - r\omega_w \searrow 0$